

DEMOGRAPHIC INFORMATION

PATIENT

NAME	[REDACTED]
DATE OF BIRTH	[REDACTED]
SEX	[REDACTED]
MEDICAL RECORD #	115100029
ACCESSION #	[REDACTED]
LAB NUMBER	[REDACTED]
FAMILY NUMBER	2370550

TEST INFORMATION

TEST NAME	Trio WGS
TEST CODE	1800
SAMPLE TYPE	[REDACTED]
DATE COLLECTED	08/15/2025
DATE RECEIVED	08/15/2025
DATE REPORTED	08/27/2025

RECIPIENT

PHYSICIAN NAME	Marissa Starr
FACILITY	CCMT
LOCATION	
PHONE	817-335-8277
FAX	817-335-8277

CLINICAL INDICATION

Based on the submitted clinical information, the patient has nystagmus, fair hair, generalized hypopigmentation, nevus flammeus, tachypnea, neonatal respiratory distress, ankyloglossia, and thick oral frenulum.

We have also received a sample from the father (DNA#3437587), the mother (DNA#3437585), and a sibling (DNA#3437531) of this individual.

RESULTS



DISEASE	INHERITANCE PATTERN	GENE/ VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
Oculocutaneous Albinism, Type IA And IB	Autosomal Recessive	TYR: c.649C>T, p.R217W	Sequence Variant	Heterozygous	Mother	Pathogenic
Oculocutaneous Albinism, Type IA And IB	Autosomal Recessive	TYR: c. [575C>A;1205G>A], p. [S192Y;R402Q]	Sequence Variant	Heterozygous	Father	Likely Pathogenic

RESULTS SUMMARY

Compound heterozygous pathogenic variant and likely pathogenic haplotype in the TYR gene were detected, consistent with a diagnosis of oculocutaneous albinism, type IB.

NAME: [REDACTED]
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ADDITIONAL FINDINGS

ACMG recommended secondary findings (v3.2; PMID 37347242)



None Detected.

Other Incidental Findings



None Detected.

Research Findings



See table in "Interpretation - Research Findings" section.

RECOMMENDATIONS

- It is recommended that correlation of these findings with the clinical phenotype be performed.
- Genetic counseling is recommended.
- For sequence variants, testing is available for this individual's at risk relatives under test code 1580. Visit <https://www.baylorgenetics.com/known-familial-mutations-policy/> for further information. Please consult with the lab for available test codes for other types of variants (CNV, STR, mito).
- Contact laboratory for requests regarding reanalysis and raw data.

NAME: [REDACTED]
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INTERPRETATION - PHENOTYPE RELATED FINDINGS

DISEASE	INHERITANCE PATTERN	GENE/ VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
Oculocutaneous Albinism, Type IA And IB	Autosomal Recessive	TYR: c.649C>T, p.R217W	Sequence Variant	Heterozygous	Mother	Pathogenic
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GENE DESCRIPTION

The TYR gene encodes an enzyme that catalyzes the first two steps in the conversion of tyrosine to melanin.

DISEASE DESCRIPTION

Defects in the TYR gene may cause autosomal recessive oculocutaneous albinism, type IA (OMIM: 203100) and type IB (OMIM:606952), which is characterized by decreased or absent pigmentation in the hair, skin, and eyes, along with ophthalmologic findings such as foveal hypoplasia, misrouting of the optic nerves, severe nystagmus, and photophobia. Oculocutaneous albinism type I is divided into two subtypes: type IA, characterized by a complete lack of tyrosinase activity due to the production of an inactive enzyme, and type IB, characterized by reduced tyrosinase activity.

VARIANT DETAILS - TYR: c.649C>T, p.R217W

- (GRCh38)chr11:89178602C>T - NM_000372.5
- The c.649C>T (p.R217W) variant in the TYR gene has been described as (likely) pathogenic in ClinVar (ID: 3795).
- This variant has been previously reported in homozygous or compound heterozygous states in individuals with oculocutaneous albinism, type I (PMID: 10987646, 27734839).
- This variant has been observed with a frequency of 0.02% in the gnomAD v4 dataset.
- This missense variant occurs at a residue with moderate evolutionary conservation, and its effect is predicted to be inconclusive (REVEL = 0.588, CADD = 20.9).

VARIANT DETAILS - TYR:c.[575C>A;1205G>A], p.[S192Y;R402Q]

- (GRCh38)chr11:89178528C>A - NM_000372.5
- The c.[575C>A;1205G>A] (p.[S192Y;R402Q]) haplotype in the TYR gene has been described as likely pathogenic in ClinVar (ID: 1301878). This complex allele has been previously reported in homozygous or compound heterozygous states in individuals with oculocutaneous albinism, type IB (PMID: 28667292, 29345414, 30679655, 31719542, 35027574).
- The frequency of this haplotype has been predicted to be 1.1% to 1.9% in European populations (PMID: 24739399, 28667292, 30679655).
- Functional studies showed this haplotype has significantly reduced tyrosinase enzyme activity (PMID: 35027574).

NAME: [REDACTED]
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INTERPRETATION - RESEARCH FINDINGS

VARIANTS IN GENES WITH NO CURRENTLY KNOWN DISEASE ASSOCIATIONS

This table includes de novo, bi-allelic, or hemizygous variants with moderate to strong supporting evidence for pathogenicity based on the variant type and zygosity. These variants may occur in a gene that is not currently known to cause disease but may be associated with disease in the future. If clinically warranted, please contact Baylor Genetics for additional analysis of the variants in the table below.

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION	ADDITIONAL DATA
NA	Not Known	ANKFN1: c.292C>A, p.R98S	Sequence Variant	Heterozygous	Father	Variant Of Uncertain Significance	GnomAD: 15/780610
NA	Not Known	ANKFN1: c.1844A>G, p.Y615C	Sequence Variant	Heterozygous	Mother	Variant Of Uncertain Significance	GnomAD: 4/779454

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QUALITY METRICS

Percentage depth of coverage at 20x: **99.03%**
Mean depth of coverage: **44.59x**

METHODOLOGY

Genomic DNA was fragmented and indexed to prepare individual libraries. The libraries were purified and pooled for sequence analysis on the Illumina NovaSeq platform for paired end reads. Internal quality control measures are performed to ensure sample integrity.

Data analysis and interpretation are performed by the Baylor Genetics analytics pipeline. Data are aligned to the human reference genome build GRCh38 using the Illumina Dragen BioIT Platform. Variant calling is performed using the Illumina Dragen haplotype-based variant calling system and Illumina Dragen genome wide depth based CNV (Copy Number Variants) caller with custom modifications from Baylor Genetics. Short tandem repeat calling is performed using the Illumina Expansion Hunter tool.

The automated genetic interpretation platform from Emedgene is implemented for variant annotation and interpretation with proprietary algorithms and open-source data sets such as gnomAD, EVS (Exome Variant Server), ClinVar, and professional resources such as HGMD (Human Gene Mutation Database) Pro. The variants were interpreted according to ACMG (American College of Medical Genetics) guidelines (reference 1) and patient phenotypes.

Variants are confirmed by Sanger sequencing, specific PCR, Multiplex Ligation-dependent Probe Amplification (MLPA), or another methodology, if sequencing or copy number variants detected by NGS do not meet internal quality standards or are within highly homologous regions.

Synonymous variants, noncoding variants not affecting splicing sites, and common benign variants are excluded from interpretation unless previously reported in the literature as possibly pathogenic variants. This test may not provide detection of certain variants or portions of certain genes due to insufficient sequencing depth, local sequence characteristics, high/low genomic complexity, or the presence of closely related pseudogenes. Small deletions or insertions larger than 25bp, low-level mosaic variants, structural variants such as inversions, and/or balanced translocations may not be detected with this technology.

Variant types being analyzed include sequencing variants and CNV in nuclear genes and mitochondrial DNA (mtDNA), and short tandem repeats (STR). Uniparental disomy (UPD) analysis is done only in a trio setting. Regions of homozygosity (ROH) are reported when greater than 5 Mb. Sensitivity of detection is decreased for mtDNA sequencing variants and CNVs at heteroplasmy levels lower than 5% and 10%, respectively. Pathogenic events of short tandem repeats can be detected within the genomic regions of following genes: AFF2, AR, ATN1, ATXN1, ATXN2, ATXN3, ATXN7, ATXN10, ATXN8OS, C9orf72, CACNA1A, CNBP, CSTB, DIP2B, DMPK, FGF14, FMR1, FXN, GLS, HTT, JPH3, NOP56, NOTCH2NLC, PABPN1, PHOX2B, PPP2R2B, RFC1, TBP, and TCF4. Sensitivity of detection is reduced for borderline or incomplete penetrance alleles.

The proband report contains results of genes related to the patient's clinical phenotype. Variants relevant to the phenotype are reported as positive (pathogenic/likely pathogenic) or variants of uncertain significance (VUS). Benign and likely benign variants are not reported. This interpretation is based on the clinical information provided by the referring institution and our current understanding of the gene and variants at the time of reporting. Additional methodologies may be used to aid in variant interpretation. Results in the report should be carefully correlated to the individual's clinical findings by the referring physician.

If elected, secondary findings including pathogenic and likely pathogenic variants in genes included in the ACMG secondary findings recommendations, or other findings determined to be severe and medically significant (incidental findings), will be reported. The ACMG recommendations (ACMG SF v3.2 (reference 2)) include the following genes: ACTA2, ACTC1, ACVRL1, APC, APOB, ATP7B, BAG3, BMPR1A, BRCA1, BRCA2, BTBD9, CACNA1S, CALM1, CALM2, CALM3, CASQ2, COL3A1, DES, DSC2, DSG2, DSP, ENG, FBN1, FLNC, GAA, GLA, HFE, HNF1A, KCNH2, KCNQ1, LDLR, LMNA, MAX, MEN1, MLH1, MSH2, MSH6, MUTYH, MYBPC3, MYH11, MYH7, MYL2, MYL3, NF2, OTC, PALB2, PCSK9, PKP2, PMS2, PRKAG2, PTEN, RB1, RBM20, RET, RPE65, RYR1, RYR2, SCN5A, SDHAF2, SDHB, SDHC, SDHD, SMAD3, SMAD4, STK11, TGFBF1, TGFBF2, TMEM127, TMEM43, TNNC1, TNNT3, TNNT2, TP53, TPM1, TRDN, TSC1, TSC2, TTN, TTR, VHL, and WT1. Variants unrelated to the patient's phenotype but potentially clinically significant in genes with no known disease association will be reported if elected.

NAME: [REDACTED]
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LIMITATIONS

The interpretation of nucleotide changes is based on our current understanding of the specific genes. This interpretation may change over time as more information about these genes becomes available. Possible diagnostic errors include sample mix-ups, genetic variants that interfere with analysis, incorrect assignment of biological parentage, or other sources. Please contact a genetic counselor at Baylor Genetics if there is reason to suspect one of these sources of error.

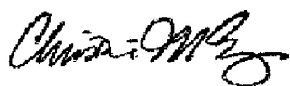
REFERENCES

1. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL; ACMG Laboratory Quality Assurance Committee. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med. 2015 May;17(5):405-24. doi: 10.1038/gim.2015.30. Epub 2015 Mar 5. PMID: 25741868; PMCID: PMC4544753.
2. Miller DT, Lee K, Abul-Husn NS, Amendola LM, Brothers K, Chung WK, Gollob MH, Gordon AS, Harrison SM, Hershberger RE, Klein TE, Richards CS, Stewart DR, Martin CL; ACMG Secondary Findings Working Group. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). Genet Med. 2023 Aug;25(8):100866. doi: 10.1016/j.gim.2023.100866. PMID: 37347242.

DISCLAIMER

This test was developed, and its performance was determined by the Baylor Miraca Genetics Laboratory DBA Baylor Genetics. It has not been cleared or approved by the U.S. Food and Drug Administration (FDA). Since FDA is not required for clinical use of this test, this laboratory has established and validated the test's accuracy and precision, pursuant to the requirement of CLIA'88. This laboratory is licensed and/or accredited under CLIA and CAP (CAP#2109314 / CLIA# 45D0660090; Lab Director: Christine M. Eng, MD).

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